

3. Design observational studies to approximate randomized trials—general advice.

3.1. *Overview.* A crucial idea when trying to estimate causal effects from an observational dataset is to conceptualize the observational dataset as having arisen from a complex randomized experiment, where the rules used to assign the treatment conditions have been lost and must be reconstructed. There are various steps that I consider essential for designing an objective observational study. These will be described in this section and then illustrated in the remaining parts of this article. In practice, the steps are not always conducted in the order given below, but often they are, especially when facing a particular candidate data set.

3.2. *What was the hypothetical randomized experiment that led to the observed dataset?* As a consequence of our conceptualization of an observational study’s data as having arisen from a hypothetical randomized experiment, the first activity is to think hard about that hypothetical experiment. To start, what exactly were the treatment conditions and what exactly were the outcome (or response) variables? Be aware that a particular observational dataset can often be conceptualized as having arisen from a variety of different hypothetical experiments with differing treatment and control conditions and possibly differing outcome variables. For example, a dataset with copious measurements of humans’ prenatal exposures to exogenous agents, such as hormones or barbiturates [e.g., [Rosenbaum and Rubin \(1985\)](#), [Reinisch et al. \(1995\)](#)], could be proposed to have arisen from a randomized experiment on prenatal hormone exposure, or a randomized experiment on prenatal barbiturate exposure, or a randomized factorial experiment on both hormone and barbiturate exposure. But the investigator must be clear about the hypothetical experiment that is to be approximated by the observational data at hand. Running regression programs is no substitute for careful thinking, and providing tables summarizing computer output is no substitute for precise writing and careful interpretation.

3.3. *Are sample sizes in the dataset adequate?* If the step presented in Section 3.1 is successful in the limited sense that measurements of both treatment conditions and outcomes seem to be available or obtainable from descriptions of the observational dataset, the next step is to decide whether the sample sizes in this dataset are large enough to learn anything of interest. Here is where traditional power calculations are relevant; also extensions, for example, involving the ratios of sample sizes needed to obtain well-matched samples [[Rubin \(1976b\)](#), Section 5], are relevant, and should be considered before plunging ahead. Sometimes, the sample sizes will be small, but the data set is the only one available to address an important question. In such

a case, it is legitimate to proceed, but efforts to create better data should be initiated.

If the available samples appear adequate, then the next step is to strip any final outcome measurements from the dataset. When designing a randomized experiment, we cannot look at any outcome measurements before doing the design, and this crucial feature of randomized experiments can be, and I believe must be, implemented when designing observational studies—outcome-free design is absolutely critical for objectivity. This point was made very strongly in Rubin (2007), but somewhat surprisingly, it was not emphasized much in older work, for example, in Cochran’s work on observational studies as reviewed in Rubin (1984), or even in most of my subsequent work on matching summarized in Rubin (2006) prior to the mid-1990s. But I now firmly believe that it is critical to hide all outcome data until the design phase is complete. A subtlety here concerns “intermediate outcome data” discussed in Section 5, such as compliance measurements.

3.4. *Who are the decision makers for treatment assignment and what measurements were available to them?* The next step is to think very carefully about why some units (e.g., medical patients) received the active treatment condition (e.g., surgery) versus the control treatment condition (e.g., no surgery): Who were the decision makers and what rules did they use? In a randomized experiment, the randomized decision rules are explicitly written down (hopefully), and in any subsequent publication, the rules are likewise typically explicitly described. But with an observational study, we have to work much harder to describe and justify the hypothetical approximating randomized assignment mechanism. In common practice with observational data, however, this step is ignored, and replaced by descriptions of the regression programs used, which is entirely inadequate. What is needed is a description of critical information in the hypothetical randomized experiment and how it corresponds to the observed data.

For example, what were the background variables measured on the experimental units that were available to those making treatment decisions, whether observed in the current dataset or not? These variables will be called the “key covariates” for this study. Was there more than one decision maker, and if so, is it plausible that all decision makers used the same rule, or nearly so, to make their treatment decisions? If not, in what ways did the decision rules possibly vary? It is remarkable to me that so many published observational studies are totally silent on how the authors think that treatment conditions were assigned, yet this is the single most crucial feature that makes their observational studies inferior to randomized experiments.

3.5. *Are key covariates measured well?* Next, consider the existence and quality of the key covariates’ measurements. If the key covariates are very

poorly measured, or not even available in the dataset being examined, it is typically a wise choice to look elsewhere for data to use to study the causal question at hand. Sometimes surrogate variables can be found that are known to be highly correlated with unmeasured key covariates and can proxy for them. But no amount of fancy analysis can salvage an inadequate data base unless there is substantial scientific knowledge to support heroic assumptions. This is a lesson that many researchers seem to have difficulty learning. Often the dataset being used is so obviously deficient with respect to key covariates that it seems as if the researcher was committed to using that dataset no matter how deficient. And interactions and nonlinear terms should not be forgotten when considering covariates that may be key; for example, the assignment rules for medical treatments could differ for those with and without medical insurance.

3.6. *Can balance be achieved on key covariates?* The next step is to try to find subgroups (subclasses, or matched pairs) of treated and control units such that within a subgroup, the treated and control units appear to be balanced with respect to their distributions of key covariates. That is, within such a subgroup, the treated and control units should look as if they could have been randomly divided (usually not with equal probability) into treatment and control conditions. Often, it will not be possible to achieve such balance in an entirely satisfactory way. In that situation, we may have to restrict inferences to a subpopulation of units where such balance can be achieved, or we may even decide that with this dataset we cannot achieve balance with enough units to make the study worthwhile. If so, we should usually forgo using this dataset to address the causal question being considered. A related issue is that if there appear to be many decision makers using differing rules (e.g., different hospitals with different rules for when to give a more expensive drug rather than a generic version), then achieving this balance will be more difficult because different efforts to create balance will be required for the differing decision makers. This point will be clearer in the context of particular examples.

3.7. *The result.* These six steps combine to make for objective observational study design in the sense that the resultant designed study can be conceptualized as a hypothetical, approximating randomized block (or paired comparison) experiment, whose blocks (or matched pairs) are our balancing groups, and where the probabilities of treatment versus control assignment may vary relatively dramatically across the blocks. This statement does not mean the researcher who follows these steps will achieve an answer similar to the one that would have been found in the analogous randomized experiment, but at least the observational study has a chance of doing so, whereas

if these steps are not followed, I believe that it is only blind luck that could lead to a similar answer as in the analogous randomized experiment.

Sometimes the design effort can be so extensive that a description of it, with no analyses of any outcome data, can be itself publishable. For a specific example on peer influence on smoking behaviors, see [Langenskind and Rubin \(2008\)](#).

4. Examples using propensity scores and subclassification.

4.1. *Classic example with one observed covariate.* The following very simple example is taken from Cochran (1968) classic article on subclassification in observational studies, which uses some smoking data to illustrate ideas. Let us suppose that we want to compare death rates (the outcome variable of primary interest) among smoking males in the U.S., where the treatment condition is considered cigarette smoking and the control condition is cigar and pipe smoking. There exists a very large dataset with the death rates of smoking males in the U.S., and it distinguishes between these two types of smokers. So far, so good, in that we have a dataset with Y and treatment indicators, and it is very large. Now we strip this dataset of all outcome data; no survival (i.e., Y) data are left and are held out of sight until the design phase is complete.

Next we ask (in a simple minded way, because this is only an illustrative example), who is the decision maker for treatment versus control, and what are the key covariates used to make this decision? It is relatively obvious that the main decision maker is the individual male smoker. It is also relatively obvious that the dominant covariate used to make this decision is age—most smokers start in their teens, and most start by smoking cigarettes, not pipes or cigars. Some pipe and cigar smokers start in college, but many start later in life. Cigarette smokers tend to have a more uniform distribution of ages. Other possible candidate key covariates are education, socio-economic status, occupational status, income, and so forth, all of which tend to be correlated with age, so to illustrate, we focus on age as our only X variable. Then our hypothetical randomized experiment starts with male smokers and randomly assigns them to cigarette or cigar/pipe smoking, where the propensity to be a cigarette smoker rather than a cigar/pipe smoker is viewed as a function of age. In this dataset, age is very well-measured. When we compare the age distribution of cigarette smokers and age distribution of cigar/pipe smokers in the U.S. in this dataset, we see that the former are younger, but that there is substantial overlap in the distributions. Before moving on to the next step, we should worry about how people in the hypothetical experiment who died prior to the assembling of the observational dataset are represented, but, for simplicity in this illustrative example, we will move on to the next step.